



Cybercrime and Cyberattacks

Objective:

Explain the effects of various cybercrimes in the IT industry, and the importance of the types of cybersecurity as protection against cyberattacks.

Identify types of Cybercrimes and perpetrators

What is cybercrime?

Cybercrime *"also called **computer crime**, the use of a computer as an instrument to further illegal ends, such as committing fraud, trafficking in child pornography and intellectual property, stealing identities, or violating privacy"*

Cyberattack

Cyberattack is *"an attempt to gain illegal access to a computer system for the purpose or causing damage or harm"* .

Types of Cyberattack

Cyberattack is defined as *"an attack launched from one or more computers against another computer, multiple computers or networks"*.

There are two (2) types of attacks that can happen:

- (1) the goal is to disable the computer or make it go off the grid;
- (2) gain access to the target's data and in most cases, control the user's computer (gain admin privileges).

Eight (8) types of Cyberattacks (Fruhlinger, 2020):

- Malware
- Phishing
- Ransomware
- Denial-of-service
- Man in the middle
- Cryptojacking
- SQL injection
- Zero-day exploits

Malware

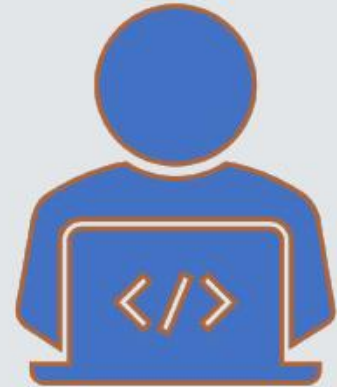
Malware refers to software that damages devices, steals data, and causes chaos. There are many types of **malware** — viruses, Trojans, spyware, **ransomware**, and more.

Virus

A type of malicious software that is loaded onto the user's computer without his/her knowledge.

It can perform malicious actions such as corrupting your file, hiding your file or even destroying your computer's hardware.

A computer virus attaches itself to a file or a program and spreads from computer to computer leaving infections as it travels.



Worms

Worms don't need other programs they propel themselves through the Internet.

Worms spread from computer to computer and has the capability to self-replicate without human intervention.

Because of its ability to self-replicate, it can deplete system resources such as memory, hard disk space and bandwidth, thus, causing your computer system to slow down.

Trojan Horse

A Trojan horse appears to be a legitimate program designed to gain access to a computer without the user's knowledge thereby getting access on important files on the user's computer.

Bots / Botnets

A malicious bot is self-propagating. It infects a host computer and connects back to its control host. The control host has now control over the infected computer. These infected computers are now called botnets and can be used for various cybercrime attacks such as Denial-of-Service (DoS) Attack.

Phishing

- **Phishing** is a cybercrime in which scammers attempt to obtain sensitive personal information such as username, passwords, account number, credit card number and the like from the target or targets by posing as someone from a legitimate institution.
- In Phishing, the victim is sent a communication that pretends to be from an official website of a reputable company. The victim is then lured into opening a website that is routed to the perpetrators.

Shared Features of Phishing Emails (Phishing.org)

- 1. Too Good To Be True** – Scammers might offer unbelievable offers that are intended to attract the target's attention.
- 2. Sense of Urgency** – In some cases, emails seem to look like from a reputable bank and telling you that they are currently updating their system and that your account will be deleted if you don't update your personal information immediately.
- 3. Hyperlinks** – In Phishing, perpetrators will provide a link that looks almost the same with the legitimate site. However, if you examine carefully, they might replace some letters like m for n.

Shared Features of Phishing Emails

(Phishing.org)

4. Attachments - If an email is sent to you with attachments from someone or from a company you do not recognize, do not open it. It might contain malicious software that might harm your computer.

5. Unfamiliar Sender – When the sender does not seem like someone you know or unfamiliar to you, do not open it.

From: Amazon <management@mazoncanada.ca> on behalf of not an Amazon email address (note the missing A in Amazon)
To: @sheridanc.on.ca
Cc:
Subject: Suspension

amazon.com

Dear Client,

Generic non-personalized greeting

We have sent you this e-mail, because we have strong reason to believe, your account has been used by someone else. In order to prevent any fraudulent activity from occurring we are required to open an investigation into this matter. We've locked your Amazon account, and you have 36 hours to verify it, or we have the right to terminate it.

To confirm your identity with us click the link below:

<https://www.amazon.com/exec/obidos/sign-in.html>

Sincerely,

Hovering over the link reveals it points to a non-Amazon site - "http://redirect.kereskedj.com"

The Amazon Associates Team



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Ransomware

Ransomware is another form of malicious software that prevents the user from accessing his/her files by encrypting it.

The attacker then demands “ransom payment” from the victim to decrypt the files and regain access.

Users are provided with instructions on how to deliver the “ransom payment” in exchange of the decryption key

How Ransomware works

One common delivery of ransomware is through email, usually done through Phishing.

Once the victim opens the link, the perpetrator can now access the victim's computer.

Perpetrators can gain administrative access on the victim's computer using social engineering tools.



Your personal files are encrypted by CTB-Locker.

Your documents, photos, databases and other important files have been encrypted with strongest encryption and unique key, generated for this computer.

Private decryption key is stored on a secret Internet server and nobody can decrypt your files until you pay and obtain the private key.

You only have 96 hours to submit the payment. If you do not send money within provided time, all your files will be permanently crypted and no one will be able to recover them.

Press 'View' to view the list of files that have been encrypted.

Press 'Next' for the next page.



WARNING! DO NOT TRY TO GET RID OF THE PROGRAM YOURSELF. ANY ACTION TAKEN WILL RESULT IN DECRYPTION KEY BEING DESTROYED. YOU WILL LOSE YOUR FILES FOREVER. ONLY WAY TO KEEP YOUR FILES IS TO FOLLOW THE INSTRUCTION.

View

95:59:59

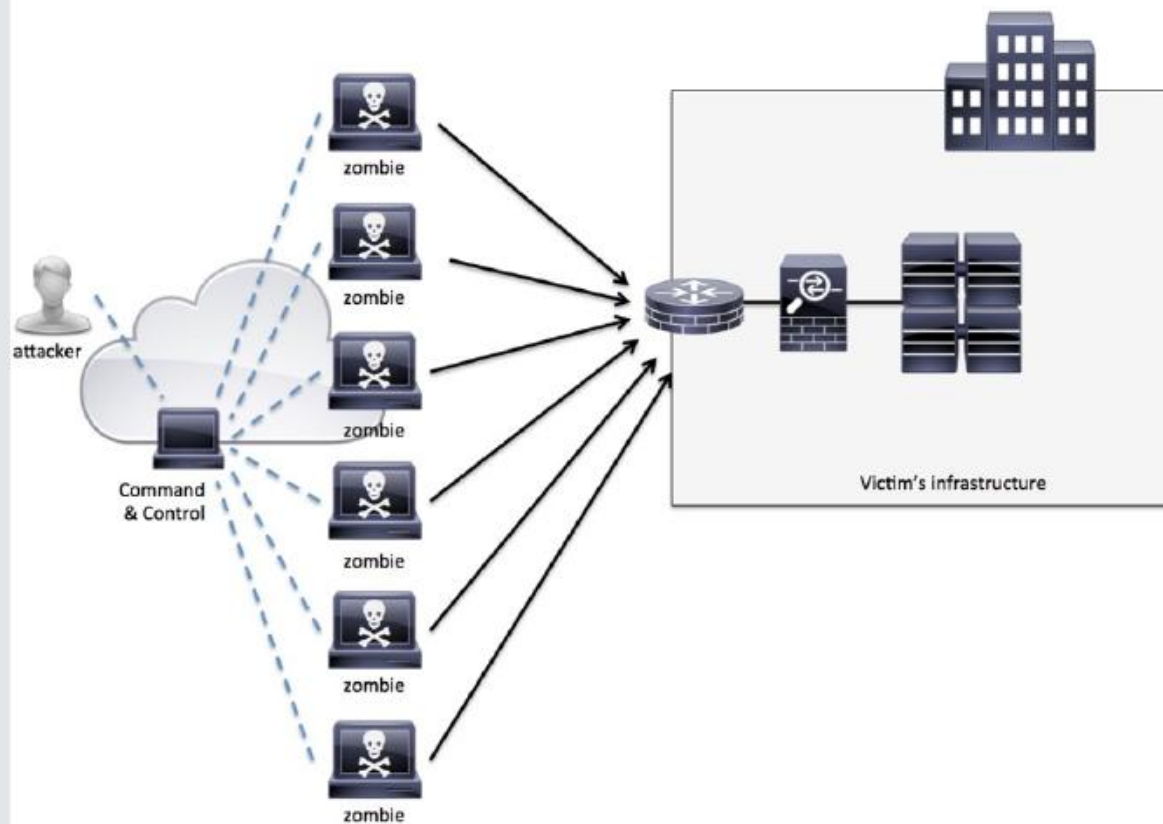
Next >>

Best Security Practices to prevent Ransomware attack

1. Ensure frequent update of your computer's operating system. Updates are important as they provide patches to security holes making your computer less vulnerable to cyber-attacks.
2. Never install software or provide administrative privileges to it unless you know what is and what it does.
3. Install an anti-virus software and keep it updated as it can detect malicious programs such as ransomware as early as the moment it enters your computer. In addition, install a whitelisting software as it can prevent unauthorized applications from running.
4. Frequently back-up your files using a cloud or external storage. In case, you become the victim, the attack would be insignificant as you have a backup copy of your files.

Denial of Service Attack

According to the website paloaltonetworks.com, a **Denial-of-Service (DoS)** attack is *“an attack meant to shut down a machine or network, making it inaccessible to its intended users. DoS attacks accomplish this by flooding the target with traffic, or sending it information that triggers a crash”*.



Denial of Service Attack

It denies the legitimate users from accessing the websites depriving them of the services or resources they expect to get. Banks and high-profile organizations are the common victims of this attack.

Denial of Service Attack

In 2000, Michael Calce (photo above), a 15-year-old boy who used the online name “Mafiaboy,” launched one of the first recorded DDoS attacks. Calce hacked into the computer networks of a number of universities. He used their servers to operate a DDoS attack that crashed several major websites, including CNN, E-Trade, eBay, and Yahoo.

Calce was convicted of his crimes in the Montreal Youth Court. As an adult, he became a “white-hat hacker” identifying vulnerabilities in the computer systems of major companies.



Michael Calce



Victims of DOS Attack

In 2016, Dyn, a major domain name system provider — or DNS — was hit with a massive DDoS attack that took down major websites and services, including AirBnB, CNN, Netflix, PayPal, Spotify, Visa, Amazon, The New York Times, Reddit, and GitHub.

The gaming industry has also been a target of DDoS attacks, along with software and media companies

Man-in-the-Middle (MitM) Attack

A man-in-the-middle attack is a type of cyberattack where a malicious actor inserts him/herself into a conversation between two parties, impersonates both parties and gains access to information that the two parties were trying to send to each other.

Man-in-the-Middle (MitM) Attack

A man-in-the-middle attack allows a malicious actor to intercept, send and receive data meant for someone else, or not meant to be sent at all, without either outside party knowing until it is too late (DuPaul, n.d.).

A Man-in-the-Middle attack is another type of cyberattack where the malicious person intercepts or eavesdrop in

Man-in-the-Middle Attack Example

Jack
Victim 1

Peter
Man in the
Middle

Jill
Victim 2

Send over your key

Send over your key

Peter sends his
own key to Jack

Jill sends her key to Jack

Jack sends his account
number as 123456789

Peter sends Jill his account
number 987654321

Jill sends money to
the wrong account

The MITM
attack
is complete

Cryptojacking

Cryptojacking is the unauthorized use of someone else's computer which may include tablets, laptops or desktop computers to mine cryptocurrency.

Cryptojacking

A common way of cryptojacking is by either getting the victim to click on a malicious link in an email that loads **cryptomining** code on the computer, or by infecting a website or online ad with JavaScript code that auto-executes once loaded in the victim's browser.

SQL Injection

SQL injection (SQLi) is a type of attack where a malicious SQL statements will be inserted and executed on data-driven applications (database servers) to manipulate data stored in any type of SQL databases.

SQL Injection

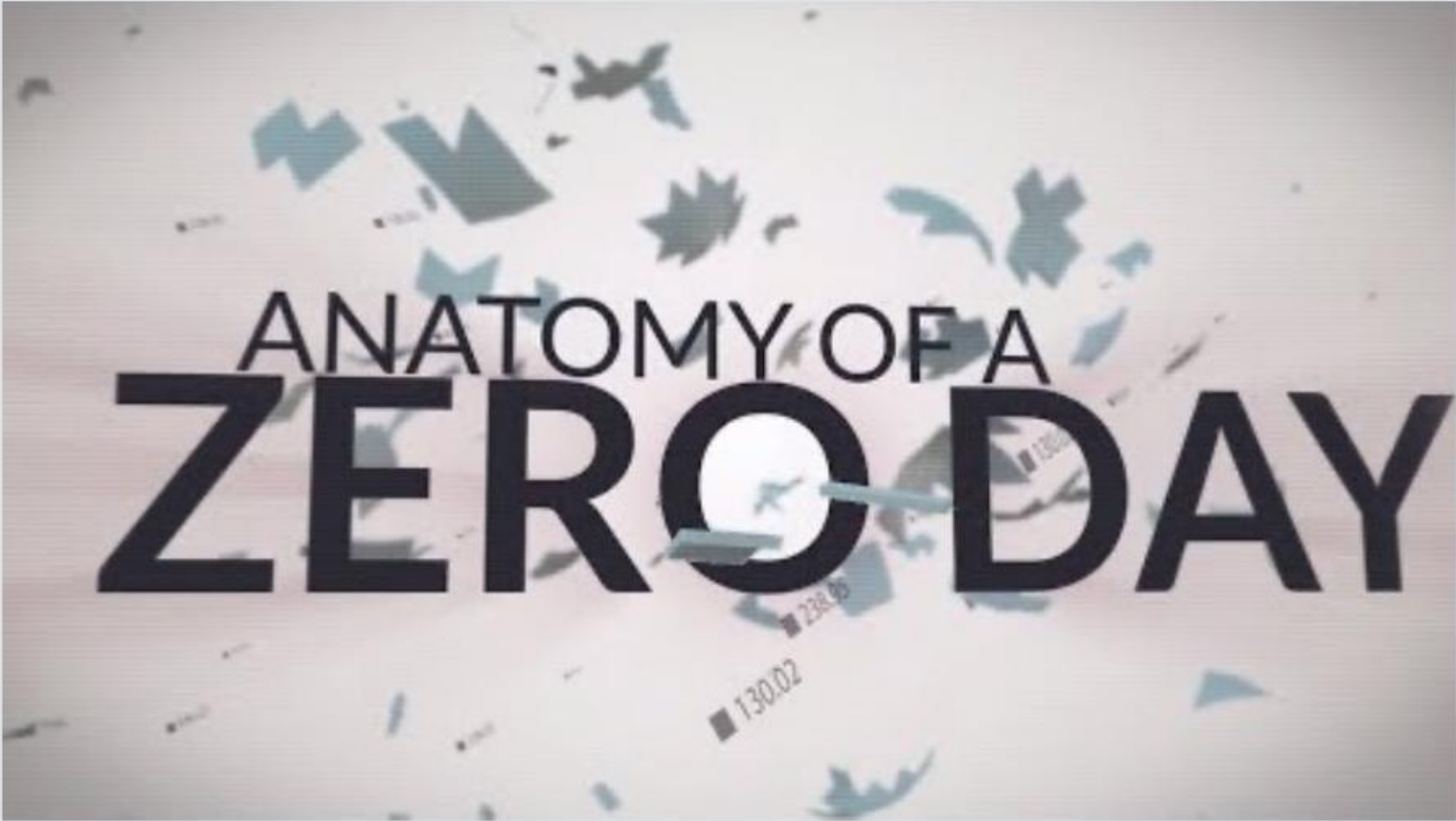
SQL injection attacks allow attackers to bypass application security measures. They can go around authentication and authorization of a web page or web application and retrieve the content of the entire SQL database. They can also use SQL Injection to add, modify, and delete records in the database

Zero Day Exploits

According to Kaspersky.com website, a **zero day exploit** is “a cyberattack that occurs on the same day a weakness is discovered in software. At that point, it's exploited before a fix becomes available from its creator”.

Zero Day Exploits

When a vulnerability or security risk is discovered by a user, the user would initially report this to the software developers. The developers would then create a patch or update to fix the error or flaw.

The background of the image is a light, off-white color. It is populated with numerous small, irregular blue fragments that resemble torn pieces of paper or confetti, scattered across the frame. A faint, light-colored world map is visible in the background, centered behind the text. The map shows the outlines of continents.

ANATOMY OF A **ZERO DAY**

Types of Cybercriminals

- Hackers
- Crackers
- Hacktivist
- Cyberterrorist
- Industrial spies

Hackers

As defined under Internet Users' Glossary under RFC 1392, a **hacker** is *"A person who delights in having an intimate understanding of the internal workings of a system, computers and computer networks in particular. The term is often misused in a pejorative context, where 'cracker' would be the correct term."*

Hackers

Hackers use their knowledge and skills in programming, and general computer security to study how the computer system and networks work and find flaws, or security issues in it. They help improve and patch vulnerabilities within the computer system and networks.

Hackers are often referred to as “white hat” or “ethical hackers”. They are what we call “the good guys”.

OH Money



Crackers

As defined under Internet Users' Glossary under RFC 1392, a cracker *"is an individual who attempts to access computer systems without authorization. These individuals are often malicious, as opposed to hackers, and have many means at their disposal for breaking into a system."*

Crackers

Crackers are also called “black hats.” They look for backdoors in programs and systems, exploit those backdoors, and steal private information for use in a malicious way.

Hacktivist

According to Oxford Online Dictionary, a **hacktivist** is defined as *"a person who gains unauthorized access to computer files or networks in order to further social or political ends"*.

A hacktivist cracks the computer system and destroys or alters data as way to express their opposition to a certain issue. Majority of the hacktivists works anonymously.

Hacktivist

*A **cyber-terrorist** is a criminal who uses computer technology and the Internet, especially to cause fear and disruption.*

Some *cyber-terrorists* spread computer viruses, and others threaten people electronically

COMELEC Website Hacked in 2016

The Anonymous Philippines hacked the website of Commission on Elections (Comelec) on March 27, 2016.

The hacktivist group defaced the website by posting a message, asking the Comelec to make sure that the machines to be used on election should have strong security features (CNN Philippines, *Anonymous PH hacks Comelec website 2016*).

Anonymous Philippines

Greetings Philippines! We are Anonymous.

The Constitution so asserts that "Sovereignty resides in the people and all government authority emanates from them."

One of the processes by which people exercise their sovereignty is through voting in election- where people choose the candidates who will best represent them and will serve them under the principle that office is a public trust."

But what happens when the electoral process is mired with questions and controversies? Can the government still guarantee that sovereignty of the people is upheld?

We request the implementation of the security features on the PCOS machines.

Commission on Elections. We are watching!

We are Anonymous.
Region.



PHOTO FROM COMELEC.GOV.PH

SAMPALOC, MANILA

GMA

COMELEC HACKER



**MIYEMBRO UMANO NG ANONYMOUS PHILIPPINES
NA NANG-HACK SA COMELEC WEBSITE, ARESTADO**

06:36 | PLOAD NA SA ISANG WEBSITE. • MATAPOS BATIKI



FORMER ANONYMOUS HACKTIVIST
JAKE DAVIS TALKS ABOUT HIS LIFE ON THE DARK NET

Industrial Spies

Industrial spies use illegal means to obtain trade secrets from competitors (Reynolds, 2019). An industrial spy may have gained an employment with the company. The primary purpose is to spy and gather information in order to obtain relevant information from the company.

An industrial spy may also be a disgruntled employee who wants to obtain revenge or use the obtained information for personal gain.

Assignment:

Please watch the documentary this documentary in Youtube:

<https://www.youtube.com/watch?v=qsKS6fwXTwg&t=23s>



Cybersecurity

What is Cybersecurity?

Cybersecurity is the practice of defending computers, servers, mobile devices, electronic systems, networks, and data from malicious attacks.

When companies become a victim of any cybercrime, client trust and reputation is at stake.

Kaspersky.com (2020) enumerates the various categories of Cyber security:

- Network security
- Application security
- Information Security
- Operational Security
- Disaster recovery and business continuity

Google Data Center Security: 6 layers



Protection against Internet Attacks

A **firewall** is hardware and/or software that protects a network's resources from intrusion by users on another network such as the Internet.

All networked and online computer users should implement a firewall solution.

Protection against Internet Attacks

A **personal firewall** is a utility program that detects and protects a personal computer and its data from unauthorized intrusions.

Some operating systems, such as Windows, include personal firewalls.

Protection against Internet Attacks

Intrusion Detection Software. Intrusion detection software automatically analyzes all network traffic, assesses system vulnerabilities, identifies any unauthorized access (intrusions), and notifies network administrators of suspicious behavior patterns or system breaches.

Firewall

